
TREE-TRANSPLANTING: AN INTER-GENERATIONAL OPERATION FOR A BALANCED AND HEALTHFUL ECOLOGY USING THE ADVANCEMENT OF TECHNOLOGY

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Grow

To grow takes time. To grow is uncertain. When a person plants a tree, he plants hope. A hope that there will be a better future. Meanwhile, to hope for a better future is to believe that growth will go with it. Unfortunately, hope, growth and belief sometimes do not turn up.

When we cut trees and plant new ones, for instance in the town of Pototan wherein an affidavit stated that 100 tree seedlings should be planted in other areas as replacement for every tree that Department of Public Works and Highways (DPWH) felled¹, did we plant hope or did we wager our growth?

Move out the way

Republic Act No. 3571 (R.A. No. 3571), enacted on June 21, 1963, prohibits the cutting, destroying or injuring of planted or growing trees, flowering plants and shrubs or plants of scenic value along public roads, in plazas, parks, school premises or in any other public pleasure ground. The exceptions for the cutting, destroying or injuring of same apply when it is necessary for public safety, or such pruning of same is necessary to enhance its beauty and only upon the recommendation of the committee and approved by the Director of Parks and Wildlife.² The law aims to prohibit the cutting, destroying, or injuring of planted or growing trees, flowering plants, shrubs, or plants of scenic value in public places. R.A. No. 3571 was amended by Presidential Decree No. 953 on July 06, 1976 providing for additional authorization on exceptions of cutting, destroying, damaging or injuring trees when it is necessary for public safety or the pruning thereof is necessary to enhance beauty, and upon the approval of the duly authorized representative of the head of agency or political subdivision having jurisdiction therein.³ These laws are as ancient as some

¹ Sornito, I. (2018, June 27), Cutting of trees in Pototan continues, Panay News, <https://www.panaynews.net/cutting-of-trees-in-pototan-continues/>

² Republic Act No. 3571 (June 21, 1963), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/25793>

³ Pres. Dec. No. 953 - Requiring the Planting of Trees in Certain Places and Penalizing Unauthorized Cutting, Destruction, Damaging and Injuring of Certain Trees, Plants and Vegetation. - Supreme Court E-Library. (1976, July 6), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/26/18327><https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/26/18327>

of the affected trees along the road during road widening. Thus, it is imperative that these legislations be reviewed to address current environmental issues.

Our Supreme Court upheld the concept of inter-generational responsibility in the case of *Oposa vs. Factoran*.⁴ The Court emphasized that every generation has a responsibility to the next to preserve that rhythm and harmony for the full enjoyment of a balanced and healthful ecology.⁵ In the field of Public International Law, 1972 Stockholm Declaration⁶ as further affirmed by 1992 Rio Declaration⁷, respectively established the inter-generational responsibility of States. The capacity of the earth to produce vital renewable resources should be maintained, and where practicable, be restored, or improved in order to guard such resources against the danger of their future exhaustion.⁸

DPWH was given permit by the Department of Environment and Natural Resources (DENR) to cut over 5,000 trees to give way to the project that covers thirteen (13) barangays in Pototan, requiring DPWH to replace each tree with at least 100 seedlings.⁹ More than 5,000 trees would be cut down to widen the two-lane national highway that links Iloilo province to the provinces of Aklan and Capiz.¹⁰ This not only happens in developing provinces but to developed provinces like Cebu Province. In cutting of roadside trees in the cities of Naga and Carcar in the province

⁴ *Oposa v. Factoran*, G.R. No. 101083, (July 30, 1993) (Phil.).

⁵ *Id.*

⁶ United Nations. (1972). United Nations Conference on the Human Environment, Stockholm 1972 | United Nations, <https://www.un.org/en/conferences/environment/stockholm1972> [Principle 2: The natural resources of the earth, including the air, water, land, flora and fauna and especially representative samples of natural ecosystems, must be safeguarded for the benefit of present and future generations through careful planning or management, as appropriate.].

⁷ United Nations. (1992). United Nations Conference on Environment and Development, Rio de Janeiro, Brazil, 3-14 June 1992 | United Nations, <https://www.un.org/en/conferences/environment/rio1992> [Principle 3: The right to development must be fulfilled so as to equitably meet developmental and environmental needs of present and future generations.].

⁸ Nachura, A. E. B., & Gatdula, J. Outline Introduction to Public International Law (2022 Edition, p. 198). REX Book Store.

⁹ The Department of Public Works and Highways (DPWH) 2nd Iloilo District Engineering Office and the Department of Environment and Natural Resources (DENR) Region 6, media briefing on the replacement of trees affected by the road widening project in Pototan, Iloilo. (May 17, 2018), <https://www.dpwh.gov.ph/dpwh/news/13277#:~:text=DPWH%20was%20given%20permit%20by,With%20at%20least%20100%20seedlings>

¹⁰ Burgos, N. P., Jr. (2016, December 1). Not 300, but 5,000 trees to be cut for road | Inquirer News. INQUIRER.net, <https://newsinfo.inquirer.net/849500/not-300-but-5000-trees-to-be-cut-for-road>

of Cebu, DENR stressed that public safety is a major consideration in granting tree-cutting permits.¹¹

Cutting trees will continue as long as there is a need to widen highways for economic transportation purposes. In the United States of America, the benefits of a transportation investment are typically estimated by comparing the amount of travel time, vehicle miles traveled and expected number of crashes for the Alternative to the Base Case.¹² As applied in Philippines highways, national agencies also aim for travel-time savings, vehicle cost operating savings, and safety benefits. Further, in the Philippine Development Plan, investments in infrastructure are linked to job creation, inclusive growth, the wider business enabling environment, and many government priorities.¹³ This shows that the government is concerned about the traffic in roads and even in highways thus they see that expansion and building new ones are the solutions available in addressing them.

Technology has advanced with development and discovery of new machineries, methods, and strategies. Hence, we should ask ourselves whether cutting trees and planting new ones is still ideal. When we cut trees, it is as if we are cutting our source of life. With the idea of planting new trees for every tree we felled, how long can we hold our breath?

Survival

The Department of Environment and Natural Resources (DENR) former chief, Secretary Roy Cimatu, sought ways to raise NGP seedlings' survival rate.¹⁴ The survival rate of its National Greening Program (NGP) seedlings in respective regions way back his tenure ranges between 50 percent to 80 percent.¹⁵ The DENR reported that there was a 78% survival rate of the trees planted under NGP from 2011 to 2016.¹⁶ An experiment showed results with loss of 20% of saplings in the first few

¹¹ Department of Environment and Natural Resources (DENR). (2024).. DENR: Public safety is a major consideration in granting tree-cutting permit, <https://denr.gov.ph/news-events/denr-public-safety-is-a-major-cons/d.eration-in-granting-tree-cutting-permit/>

¹² Benefit-Cost Analysis - MNDOT. <https://www.dot.state.mn.us/planning/program/benefitcost.html>

¹³ Patinio, F. (2018). DPWH to build more roads to ease Metro Manila traffic. <https://www.pna.gov.ph/articles/1045213>

¹⁴ Teves, C. (2018). DENR chief seeks ways to raise NGP seedlings' survival rate. <https://www.pna.gov.ph/articles/1056443>

¹⁵ *Id.*

¹⁶ Gozum, I. (2023, September 26). Lawmaker seeks bigger maintenance, compensation budget for reforestation, <https://pid.s.gov.ph/details/news/in-the-news/lawmaker-seeks-bigger-maintenance-compensation-budget-for-reforestation#:~:text=The%20Department%20of%20Environment%20and,NGP%20from%202011%20to%202016>

months amounted to approximately 4800 plants, and approximately 12,000 plants in the first 20 months.¹⁷

While the result shows a promising rate, it is uncertain whether the seedling will prosper into a full-grown tree. It is highly likely that the rate will go down considering rapid changes in our climate. In Calasiao, Pangasinan, at least 2,500 saplings and seedlings of native trees in the mountains of Mangatarem town in this province did not survive the recent heat and have withered from extremely high temperatures.¹⁸

Tree of spades

As a tool, tree spades have been around in some form since as early as 1961, and make moving trees only possible but far less laborious.¹⁹ It consists of a series of sharp blades that are attached to a hydraulic system which are used to dig around the roots of a tree, creating a root ball.²⁰ The root ball is then lifted out of the ground and moved to a new location.²¹ With the advancement of mechanical engineering, a tree spade now comes with a variety of sizes attached to heavy-duty trucks or construction vehicles.

Transplanting is the term used to describe the digging and replanting of trees from one location to a new location hence, due to the wide extent of morphology of the tree root system, transplanting of trees involves substantial removal of roots.²² Transplanting or moving trees or shrubs from one location to another site is a major operation from which most plants recover slowly.²³ There are numerous benefits to using a tree spade than a chainsaw. Aside from saving time and efforts, using tree spade enables us to transport mature trees at a higher survival rate that would be hard to move otherwise, preserve trees that would otherwise be destroyed, save labor costs

¹⁷ Preece, N. D., Van Oosterzee, P., & Lawes, M. J. (2023). Reforestation success can be enhanced by improving tree planting methods. *Journal of Environmental Management*, 336, 117645, <https://doi.org/10.1016/j.jenvman.2023.117645>

¹⁸ Sotelo, Y. (2024, April 19). Intense heat takes toll on young native trees in Pangasinan. INQUIRER.net, <https://newsinfo.inquirer.net/1931222/intense-heat-takes-toll-on-young-native-trees-in-pangasinan>

¹⁹ Campbell, R. (2023, December 28). Using a tree spade to preserve your favorite tree. Miller's Tree Service, <https://www.millertreesrv.com/how-to-use-a-tree-spade/>

²⁰ Insta-Shade, O. (n.d.). Tree Spades - Oxford Insta-Shade. Oxford Insta-Shade. <https://www.oxfordinstashade.com/tree-spades>

²¹ *Id.*

²² Greening, Landscape and Tree Management Section Development Bureau. (2014). Guidelines on Tree Transplanting, https://www.greening.gov.hk/filemanager/greening/en/content_28/Guidelines_on_Tree_Transplanting_e.pdf

²³ Transplanting trees and shrubs. (n.d.). Natural Resource Stewardship, <https://naturalresources.extension.iastate.edu/encyclopedia/transplanting-trees-and-shrubs>

required for hand-digging and removal, and by giving us the ability to move established trees to landscape an area, rather than waiting for saplings to grow.²⁴

Planting an Idea

We should invest in tree spade as it has proven to offer mutual benefits between us and the environment. During the replacing of trees felled to widen the national highway in Pototan town, DPWH acknowledged it has been slow in replacing the said felled trees. According to Edna Mae Cañete, assistant engineer of DPWH-6's 2nd Engineering District during the duration of the project, DPWH did not have enough funds to procure tree seedlings as the budget for the highway widening did not include the needed money for tree planting.²⁵ Applying the most efficient strategy and logistics, several tree spades are enough for an entire region as it is not always on demand unlike other heavy construction machinery. Another obstacle encountered during the cutting down of trees was the lack of expertise to plant tree seedlings requiring DENR's rigorous assistance.

There is no guarantee that a tree planted today will be alive in the upcoming years. Hence, the best practice is to allow fully grown trees which are thriving at the moment to continue with its purpose. A study of tree growth, mortality and harvest at plantations in the Philippines shows that the growth rate of individual trees differed widely in the mixed plantations, ranging from 0.001 to 271.44 cm²/year.²⁶ This is not to discourage planting trees but to prioritize an idea that it is better to transplant than to cut then plant. There is no harm in pursuing this policy of using the advancement of technology to save a living thing that is not yet possible to recreate. We can always buy seedlings, but nowhere will we be able to buy the time it takes for trees to grow.

²⁴ Heritage. (2024, April 4). What's a tree spade – and can you actually dig up a tree with it? - Heritage Tree Care. Heritage Tree Care, <https://www.heritagetreecare.com.au/whats-a-tree-spade-and-can-you-actually-dig-up-a-tree-with-it/>

²⁵ Sornito, I. (2018, May 14).. DPWH slow to replace felled trees in Pototan. Panay News, <https://www.panaynews.net/dpwh-slow-to-replace-felled-trees-in-pototan/>

²⁶ Nguyen, H., Vanclay, J., Herbohn, J., & Firm, J. (2016). Drivers of Tree Growth, Mortality and Harvest Preferences in Species-Rich Plantations for Smallholders and Communities in the Tropics, <https://doi.org/10.1371/journal.pone.0164957>

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